# SYNTH-06: Synthetic Analytics & Alerting

> **Series:** SYNTH — Synthetic Monitoring | **Notebook:** 6 of 6 | **Created:** December 2025 | **Last Updated:** 04/25/2026

## Dashboards, SLOs, and Alerting Strategies

This notebook covers advanced analytics for synthetic monitoring, including building dashboards, configuring SLOs, and implementing effective alerting strategies using the latest Dynatrace platform.

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## Prerequisites

- ✅ Access to a Dynatrace environment with Synthetic Monitoring
- ✅ Completed SYNTH-01 through SYNTH-05
- ✅ Active synthetic monitors generating data

# 1. Analytics Overview

## Key Metrics for Synthetic Monitoring

| Metric Category | Metrics | Use Case |
| --- | --- | --- |
| **Availability** | Success rate, failure count | SLA compliance |
| **Performance** | Response time, TTFB, DNS | User experience |
| **Reliability** | Consecutive failures, MTTR | Stability |
| **Geographic** | Per-location metrics | Regional issues |

## Data Sources

| Source | Description | Best For |
| --- | --- | --- |
| `bizevents` | Execution results | Real-time analysis |
| `dt.entity.synthetic_test` | Monitor definitions | Configuration |
| `dt.entity.synthetic_location` | Location info | Geographic analysis |
| `metrics` | Aggregated metrics | Long-term trends |

# 2. Availability Analysis

## Calculating Availability

```
Availability = (Successful Executions / Total Executions) × 100%

Example SLA Targets:
- 99.9% = max 8.76 hours downtime/year
- 99.5% = max 43.8 hours downtime/year
- 99.0% = max 87.6 hours downtime/year
```

```
// Overall synthetic availability (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
  }
| fieldsAdd availability_pct = round((successful * 100.0) / total_executions,
decimals: 3)
| fieldsAdd downtime_minutes = round((failed * 5.0), decimals: 0)  //
Assuming 5-min frequency
```

```
// Availability by monitor (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
  }, by: {dt.entity.synthetic_test}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
3)
| sort availability_pct asc
| limit 30
```

```
// Availability trend over time
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_count = count()
  }, by: {hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) / total_count,
decimals: 2)
| sort hour_bucket desc
```

```
// Daily availability report
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd day = formatTimestamp(timestamp, format: "yyyy-MM-dd")
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
  }, by: {day}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
3)
| sort day desc
| limit 30
```

# 3. Performance Analysis

## Performance Metrics

| Metric | Description | Good | Warning | Critical |
|---|---|---|---|---|
| **Response Time** | Total execution time | < 2s | 2-5s | > 5s |
| **DNS** | DNS resolution | < 50ms | 50-200ms | > 200ms |
| **Connect** | TCP connection | < 100ms | 100-300ms | > 300ms |
| **TTFB** | Time to first byte | < 500ms | 500ms-1s | > 1s |

```
// Response time percentiles by monitor
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    p50_ms = percentile(toDouble(synthetic.response_time), 50),
    p75_ms = percentile(toDouble(synthetic.response_time), 75),
    p90_ms = percentile(toDouble(synthetic.response_time), 90),
    p95_ms = percentile(toDouble(synthetic.response_time), 95),
    p99_ms = percentile(toDouble(synthetic.response_time), 99),
    executions = count()
  }, by: {dt.entity.synthetic_test}
| sort p95_ms desc
| limit 20
```

```
// Performance trend comparison (this week vs last week)
fetch bizevents, from: now() - 14d
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| fieldsAdd week = if(timestamp &gt; now() - 7d, "This Week", else: "Last
Week")
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
  }, by: {dt.entity.synthetic_test, week}
| sort dt.entity.synthetic_test, week
```

```
// Timing breakdown analysis
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_dns_ms = avg(toDouble(synthetic.dns_time)),
    avg_connect_ms = avg(toDouble(synthetic.connect_time)),
    avg_ssl_ms = avg(toDouble(synthetic.ssl_time)),
    avg_ttfb_ms = avg(toDouble(synthetic.time_to_first_byte)),
    avg_download_ms = avg(toDouble(synthetic.download_time)),
    avg_total_ms = avg(toDouble(synthetic.response_time))
  }, by: {dt.entity.synthetic_test}
| fieldsAdd other_ms = avg_total_ms - avg_dns_ms - avg_connect_ms -
avg_ssl_ms - avg_ttfb_ms - avg_download_ms
| sort avg_total_ms desc
| limit 20
```

```
// Performance anomalies (response time &gt; 2x average)
// First calculate average per monitor, then find executions above threshold
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| fieldsAdd response_ms = toDouble(synthetic.response_time)
| summarize {
    avg_response = avg(response_ms),
    max_response = max(response_ms),
    min_response = min(response_ms),
    executions = count()
  }, by: {dt.entity.synthetic_test}
| fieldsAdd deviation_factor = round(max_response / avg_response, decimals: 1)
| filter deviation_factor &gt; 2
| sort deviation_factor desc
| limit 50
```

# 4. Location Comparison

## Geographic Analysis

Compare synthetic results across locations to:
- Identify regional performance issues
- Detect CDN or DNS problems
- Validate global availability
- Optimize user experience by region

```
// Availability by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
  }, by: {dt.entity.synthetic_location}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals: 2)
| sort availability_pct asc
| limit 30
```

```
// Performance by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p50_ms = percentile(toDouble(synthetic.response_time), 50),
    p95_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
  }, by: {dt.entity.synthetic_location}
| sort avg_response_ms desc
| limit 30
```

```
// Location performance heatmap (monitor x location)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    availability_pct = countIf(synthetic.availability == true) * 100.0 /
count()
  }, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| sort dt.entity.synthetic_test, avg_response_ms desc
```

```
// Location-specific failures
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    unique_monitors = countDistinct(dt.entity.synthetic_test),
    error_types = collectDistinct(synthetic.error_message)
  }, by: {dt.entity.synthetic_location}
| sort failure_count desc
| limit 20
```

# 5. SLO Configuration

## Service Level Objectives for Synthetic

| SLO Type | Metric | Example Target |
|---|---|---|
| **Availability** | Success rate | 99.9% |
| **Performance** | Response time P95 | < 3000ms |
| **Error Budget** | Allowed failures | 0.1% of executions |

## Creating Synthetic SLOs

1. **Navigate to**: Settings → Service-level objectives
2. **Create SLO**: Name, description, timeframe
3. **Define metric**: Use synthetic availability or response time
4. **Set target**: e.g., 99.9% availability
5. **Configure alerting**: Error budget alerts

```
// SLO calculation - Availability (30-day window)
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
  }, by: {dt.entity.synthetic_test}
| fieldsAdd sli_availability = round((successful * 100.0) / total_executions,
decimals: 4)
| fieldsAdd slo_target = 99.9
| fieldsAdd error_budget_total = round(total_executions * 0.001, decimals: 0)
// 0.1% budget
| fieldsAdd error_budget_remaining = round(error_budget_total - failed,
decimals: 0)
| fieldsAdd error_budget_pct = round((error_budget_remaining * 100.0) /
error_budget_total, decimals: 1)
| fieldsAdd slo_status = if(sli_availability &gt;= slo_target, "✅ Met",
else: "❌ Breached")
| sort sli_availability asc
| limit 30
```

```
// SLO calculation - Performance (P95 &lt; 3000ms)
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    executions = count(),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    within_target = countIf(toDouble(synthetic.response_time) &lt; 3000)
  }, by: {dt.entity.synthetic_test}
| fieldsAdd sli_performance = round((within_target * 100.0) / executions,
decimals: 2)
| fieldsAdd slo_target = 95.0  // 95% of requests &lt; 3s
| fieldsAdd slo_status = if(sli_performance &gt;= slo_target, "✅ Met", else:
"❌ Breached")
| sort sli_performance asc
| limit 30
```

```
// Error budget burn rate (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd day = formatTimestamp(timestamp, format: "yyyy-MM-dd")
| summarize {
    total = count(),
    failed = countIf(synthetic.availability == false)
  }, by: {dt.entity.synthetic_test, day}
| fieldsAdd daily_error_rate = round((failed * 100.0) / total, decimals: 3)
| fieldsAdd budget_consumed = round(failed * 100.0 / (total * 0.001),
decimals: 1)  // vs 0.1% budget
| sort dt.entity.synthetic_test, day desc
```

# 6. Alerting Strategies

## Alert Types

| Alert Type | Trigger | Use Case |
|---|---|---|
| **Availability** | Consecutive failures | Outage detection |
| **Performance** | Response time threshold | Degradation |
| **SLO** | Error budget consumption | Proactive warning |
| **SSL** | Certificate expiration | Security |

## Alert Configuration Best Practices

| Setting | Recommendation | Reason |
| --- | --- | --- |
| **Consecutive failures** | 2-3 | Avoid false positives |
| **Location threshold** | 2+ locations | Confirm not local issue |
| **Alert delay** | 5-10 minutes | Allow for transients |
| **Auto-resolve** | After 2 successes | Clear resolved issues |

```
// Monitors with recent failures (potential outages)
// Check for monitors that have failed multiple times recently
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    last_failure = max(timestamp),
    first_failure = min(timestamp)
  }, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| filter failure_count &gt;= 2
| sort failure_count desc
| limit 20
```

```
// Performance degradation alerts (P95 &gt; threshold)
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
  }, by: {dt.entity.synthetic_test}
| filter p95_response_ms &gt; 5000  // &gt; 5 seconds threshold
| sort p95_response_ms desc
```

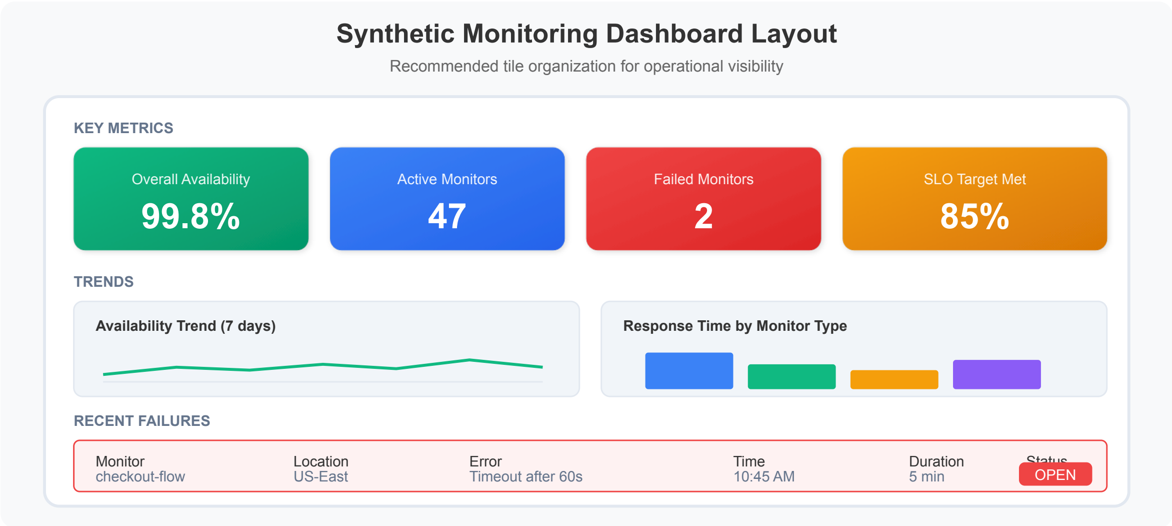
```
// Multi-location failure detection (global outage indicator)
fetch bizevents, from: now() - 30m
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failing_locations = countDistinct(dt.entity.synthetic_location),
    failure_count = count(),
    locations = collectDistinct(dt.entity.synthetic_location)
  }, by: {dt.entity.synthetic_test}
| filter failing_locations &gt;= 2
| fieldsAdd severity = if(failing_locations &gt;= 3, "CRITICAL", else:
"WARNING")
| sort failing_locations desc
```

# 7. Dashboard Building

## Recommended Dashboard Tiles

| Tile Type | Content | Visualization |
| --- | --- | --- |
| **Overall Health** | Availability % | Single value |
| **Availability Trend** | Hourly availability | Line chart |
| **Response Time** | P95 by monitor | Bar chart |
| **Location Map** | Geographic availability | World map |
| **Failures Table** | Recent failures | Table |
| **SLO Status** | Error budget | Gauge |

# Dashboard Layout




```
// Dashboard tile: Overall availability (single value)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    availability_pct = round(countIf(synthetic.availability == true) * 100.0
/ count(), decimals: 2)
  }
```

```
// Dashboard tile: Active vs failed monitors
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    latest_status = takeFirst(synthetic.availability)
  }, by: {dt.entity.synthetic_test}
| summarize {
    total_monitors = count(),
    healthy = countIf(latest_status == true),
    failing = countIf(latest_status == false)
  }
```

```
// Dashboard tile: Response time by monitor (bar chart)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95)
  }, by: {dt.entity.synthetic_test}
| sort p95_response_ms desc
| limit 10
```

```
// Dashboard tile: Recent failures table
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| fields timestamp,
         monitor = dt.entity.synthetic_test,
         location = dt.entity.synthetic_location,
         error = synthetic.error_message
| sort timestamp desc
| limit 20
```

```
// Dashboard tile: Hourly availability trend (line chart)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_executions = count(),
    failures = countIf(synthetic.availability == false)
  }, by: {hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) /
total_executions, decimals: 2)
| sort hour_bucket asc
```

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## Summary

In this notebook, you learned:

✅ **Availability analysis** - Calculating and trending availability
✅ **Performance analysis** - Percentiles, timing breakdown, anomalies
✅ **Location comparison** - Geographic performance analysis

✅ **SLO configuration** - Service level objectives and error budgets
✅ **Alerting strategies** - Outage detection, performance alerts
✅ **Dashboard building** - Key tiles and layouts

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## Series Complete!

You have completed the Synthetic Monitoring Best Practices series:

1. ✅ **SYNTH-01**: Fundamentals
2. ✅ **SYNTH-02**: Browser Monitors
3. ✅ **SYNTH-03**: HTTP Monitors
4. ✅ **SYNTH-04**: Private Locations
5. ✅ **SYNTH-05**: Network Monitoring
6. ✅ **SYNTH-06**: Analytics & Alerting

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## References

- Synthetic Monitoring Overview
- Analyze Synthetic Monitors
- Service Level Objectives
- Alerting Profiles
- Dashboards

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*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*